CPSC 304 Project Cover Page

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

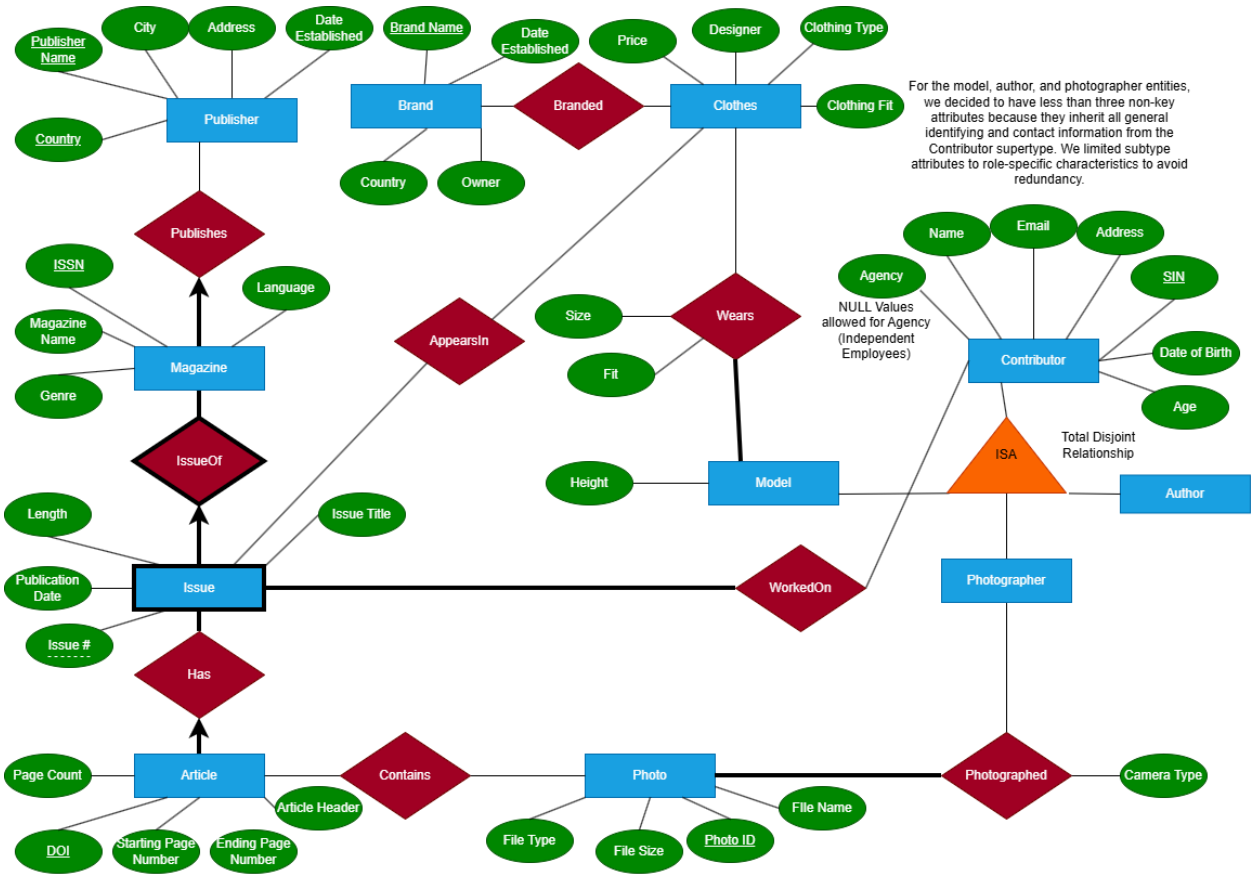
In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

2. Our database is designed for a Fashion Magazine Management system, where it models information about the magazine and its publisher. This is further broken down by individual issues (like issue number/date) and what's inside each issue (articles and their page ranges). It also keeps track of the people involved with creating the magazine, like writers, photographers, and models, plus the photos and other assets that appear in the magazine.

3. We will be using JavaScript/React, and the department-provided Oracle database.

4. We decided to apply all the suggestions that were given to us from Milestone 1. This includes:

i) Making Author an entity that is part of an ISA relationship with Contributor.
ii) Creating a new Brand entity that is in a Branded relationship with Clothing entity.
iii) Changing the ISA relationship to be a total disjoint relationship.
iv) Fulfilling the required 3 non-key attributes for each entity.
v) Making the WorkedOn relationship between the Contributor entity and the Issue, while Photographer has a new relationship with Photo, called Photographed.
vi) Having Issue # correctly labeled as the partial key for the Issue entity.
vii) Changing the primary key of Article to be DOI instead of header.



5. Primary key is underlined, notnull is **bolded**

a) Publisher(pname: char[20], country: char[2], city: char[20], address: char[30], established: date)
   - CKs: {pname, country}

Magazine(ISSN: char[8], mname: char[20], genre: char[20], language: char[20], **pname**: char[20], **country**: char[2])
   - CK: {ISSN}
   - FK: pname and country (from Publisher)

Issue(issuenum: int, issuetitle: char[20], length: int, pdate: date, ISSN: char[8])
   - CK: {issuenum, ISSN}
   - FK: ISSN (from Magazine)

Article(doi: char[20], pagecount: int, header: char[50], pagestart: int, pageend: int, **issuenum**: int, **ISSN**: char[8])
   - CK: {doi}
   - FK: issuenum and ISSN (from Issue)

Photo(filetype: char[5], filesize: int, pid: int, filename: char[20])
   - CK: {pid}

Contains(doi: char[20], pid: int)
   -CK: {doi, pid}
   -FK: doi (from Article), pid (from Photo)

Contributor(agency: char[20], cname: char[20], email: char[20], address: char[30], sin: char[9], birthday: date, age: int)
   -CK: {sin}

Author(sin: char[9])
   -CK: {sin}
   -FK: sin (from Contributor)

Photographer(sin: char[9])
   -CK: {sin}
   -FK: sin (from Contributor)

Model(sin: char[9], height: int)
   -CK: {sin}
   -FK: sin (from Contributor)

Photographed(sin: char[9], pid: int, camtype: char[20])
   -CK: {sin, pid}

-FKs: sin (from Contributor), pid (from Photo)

WorkedOn(sin: char[9], issuenum: int, ISSN: char[8])
-CK: {sin, issuenum, ISSN}
-FKs: sin (from Contributor), issuenum and ISSN (from Issue)

Clothes(price: int, designer: char[20], ctype: char[20], cfit: char[20], cid: int)
-CK: {cid}

AppearsIn(cid: int, issuenum: int, ISSN: char[8])
-CK: {cid, issuenum, ISSN}
-FKs: cid (from Clothes), issuenum and ISSN (from Issue)

Wears(sin: char[9], cid: int, size: char[3], fit: char[20])
-CK: {sin, cid}
-FKs: sin (from Contributor), cid (from Clothes)

Brand(bname: char[20], established: date, country: char[2], owner: char[20])
-CK: {bname}

Branded(bname: char[20], cid: int)
-CK: {bname, cid}
-FKs: bname (from Brand), cid (from Clothes)

6.

a) Publisher:
   i) pname, country -> city, address, established
   ii) address, city -> country
   iii) address -> city

b) Magazine:
   i) ISSN -> mname, genre, language, pname, country
   ii) mname, pname -> ISSN

c) Issue:
   i) issuenum, ISSN -> issuetitle, length, pdate
   ii) issuetitle, pdate, ISSN -> issuenum, length

d) Article:
   i) doi -> header, pagestart, pageend, pagecount, issuenum, ISSN
   ii) pagestart, pageend -> pagecount
   iii) pagestart, pagecount -> pageend
   iv) pagecount, pageend -> pagestart

v) pagestart, issuenum, ISSN -> doi

- Assume no two articles in the same issue share a starting or ending page.

e) Photo:
   i) pid -> filename, filetype, filesize
   ii) filename -> filetype

f) Contributor:
   i) sin -> agency, cname, email, address, birthday, age
   ii) email -> agency, cname, address, birthday, age
   iii) birthday -> age

g) Model:
   i) sin -> height

- In this relation, no other attribute is able to determine height and has no hidden dependencies.

h) Clothes:
   i) cid -> ctype, cfit, designer, price
   ii) designer, ctype -> price

i) Brand:
   i) bname -> established, country, owner

- In this relation, a brand's founding date, country, and owner are all direct facts about the brand itself. None of these attributes reasonably determines the others.

j) Wears:
   i) sin, cid -> size, fit

- Size and fit are specific to the model and the clothing they are wearing. There are no hidden dependencies between size and fit, as a model can wear the same size in different fits.

k) Photographed:
   i) pid, sin -> camtype

- There are no hidden dependencies, as camtype alone cannot determine other attributes in the Photographed entity.

- Author and Photographer do not have relational schemas as they are ISA subtype relations. There are no additional attributes specific to them beyond their identity as a contributor, which is why there are no functional dependencies listed.

7. We used the 3NF synthesis because it guarantees dependency preservation and a lossless decomposition. BCNF may eliminate functional dependencies.

a) Publisher:
   - pname, country -> city, address, established
   - address, city -> country
   - address -> city

   The closures are:
   {pname, country}+ = pname, country, city, address, established
   {address, city}+ = city, address, country
   {address}+ = address, city, country

   Aside from FD(pname, country -> city, address, established), the remaining FDs violate 3NF. This is because their left-hand side is not a superkey, and their right-hand side contains a non-prime attribute.

- Put FDs in Standard Form:
  - pname, country -> city
  - pname, country -> address
  - pname, country -> established
  - address, city -> country
  - address -> city

- Minimize LHS of each FD:
  - pname, country -> address
  - pname, country -> city
  - pname, country -> established
  - address -> country
  - address -> city

- Delete Redundant FDs:
  - pname, country -> address
  - pname, country -> established
  - address -> country
  - address -> city

  FD(pname, country -> city) dropped because you can derive city transitively through FD(pname, country -> address) and FD(address ->city).

- Synthesis Methods: Creating relations for every FD in the minimal cover
  - S1(<u>pname</u>: char[20], <u>country</u>: char[2], address: char[30], established: date)
  - S2(<u>address</u>: char[30], city: char[20], country: char[2])

Since S1 contains the candidate key of the original relation, no need to add an extra relation. No relation is a subset of another, so there are no redundant relations.

b) Magazine:
   - ISSN -> mname, genre, language, pname, country
   - mname, pname -> ISSN

   The closures are:
   {ISSN}+ = ISSN, mname, genre, language, pname, country
   {mname, pname}+ = mname, pname, ISSN

   All FDs have a left-hand side that is a superkey or a right-hand side that contains a prime attribute (part of a candidate key). This is in 3NF.

- Synthesis Methods: Creating relations for every FD in the minimal cover
  - S1(ISSN: char[8], mname: char[20], genre: char[20], language: char[20], pname: char[20], country: char[2])
  - S2(mname: char[20], pname: char[20], ISSN: char[8])

Since S1 contains the candidate key of the original relation, no need to add an extra relation. S2 is a subset of S1, so S2 can be removed.
  - S1(ISSN: char[8], mname: char[20], genre: char[20], language: char[20], pname: char[20], country: char[2])

c) Issue:
   - issuenum, ISSN -> issuetitle, length, pdate
   - issuetitle, pdate, ISSN -> issuenum, length

   The closures are:
   {issuenum, ISSN}+ = issuenum, ISSN, issuetitle, length, pdate
   {issuetitle, pdate, ISSN}+ = issuetitle, pdate, ISSN, issuenum, length

   The superkeys are (issuenum, ISSN) and (issuetitle, pdate, ISSN).

   All FDs have a left-hand side that is a superkey or a right-hand side that has a prime attribute (part of a candidate key). This is in 3NF.

- Synthesis Methods: Creating relations for every FD in the minimal cover
  - S1(issuenum: int, ISSN: char[8], issuetitle: char[20], length: int, pdate: date)
  - S2(issuetitle: char[20], pdate: date, ISSN: char[8], issuenum: int)

Since S1 contains the candidate key of the original relation, no need to add an extra relation. S2 is a subset of S1, so S2 can be removed.

- S1(<u>issuenum</u>: int, <u>ISSN</u>: char[8], issuetitle: char[20], length: int, pdate: date)

d) Article:

- doi -> header, pagestart, pageend, pagecount, issuenum, ISSN
- pagestart, pageend -> pagecount
- pagestart, pagecount -> pageend
- pagecount, pageend -> pagestart
- pagestart, issuenum, ISSN -> doi

The closures are:
{doi}+ = doi, header, issuenum, ISSN, pagestart, pageend, pagecount
{pagestart, pageend}+ = pagestart, pageend, pagecount
{pagestart, pagecount}+= pagestart, pagecount, pageend
{pagecount, pageend}+ = pagecount, pageend, pagestart
{pagestart, issuenum, ISSN}+ = pagestart, issuenum, ISSN, header, pagecount, pageend, doi

Aside from FD(doi -> pagestart, pageend, pagecount, issuenum, ISSN) and FD(pagestart, issuenum, ISSN -> header, pagecount, pageend, doi) the remaining FDs violate 3NF. This is because their left-hand side is not a superkey, and their right-hand side contains a non-prime attribute.

- Put FDs in Standard Form:
  - doi-> pagestart
  - doi-> pageend
  - doi -> pagecount
  - doi -> issuenum
  - doi -> ISSN
  - doi -> header
  - pagestart, pageend -> pagecount
  - pagestart, pagecount -> pageend
  - pagecount, pageend -> pagestart
  - pagestart, issuenum, ISSN -> doi

- Minimize LHS of each FD (No Change):
  - doi-> pagestart
  - doi-> pageend
  - doi -> pagecount
  - doi -> issuenum
  - doi -> ISSN
  - doi -> header

- pagestart, pageend -> pagecount
- pagestart, pagecount -> pageend
- pagecount, pageend -> pagestart
- pagestart, issuenum, ISSN -> doi

- Delete Redundant FDs:
  - doi-> pagestart
  - doi-> pageend
  - doi -> issuenum
  - doi -> ISSN
  - doi -> header
  - pagestart, pageend -> pagecount
  - pagestart, pagecount -> pageend
  - pagecount, pageend -> pagestart
  - pagestart, issuenum, ISSN -> doi

  FD(doi -> pagecount) is dropped because you can derive pagecount transitively through FD(doi -> pagestart), FD(doi -> pageend), and FD(pagestart, pageend -> pagecount).

- Synthesis Methods: Creating relations for every FD in the minimal cover
  - S1(doi: char[20], pagestart: int, pageend: int, issuenum: int, ISSN: char[8], header: char[50])
  - S2(pagestart: int, pageend: int, pagecount: int)
  - S3(pagestart: int, pagecount: int, pageend: int)
  - S4(pagecount: int, pageend: int, pagestart: int)
  - S5(pagestart: int, issuenum: int, ISSN: char[8], header: char[50])

Since S1 contains the candidate key of the original relation, no need to add an extra relation. S3 and S4 represent the same attributes as S2. S5 is a subset of S1, so they can be removed.

  - S1(doi: char[20], pagestart: int, pageend: int, issuenum: int, ISSN: char[8], header: char[50])
  - S2(pagestart: int, pageend: int, pagecount: int)

e) Photo:
  - pid -> filename, filetype, filesize
  - filename -> filetype

The closures are:
{pid}+ = pid, filename, filetype, filesize
{filename}+ = filename, filetype

Aside from FD(pid -> filename, filetype, filesize), the remaining FD violates 3NF. This is because their left-hand side is not a superkey, and their right-hand side contains a non-prime attribute.

- Put FDs in Standard Form:
    - pid -> filename
    - pid -> filetype
    - pid -> filesize
    - filename -> filetype

- Minimize LHS of each FD (No Change):
    - pid -> filename
    - pid -> filetype
    - pid -> filesize
    - filename -> filetype

- Delete Redundant FDs:
    - pid -> filename
    - pid -> filesize
    - filename -> filetype

  FD(pid -> filetype) is dropped because you can derive filetype transitively through FD(pid -> filename), and FD(filename -> filetype).

- Synthesis Methods: Creating relations for every FD in the minimal cover
    - S1(<u>pid</u>: int, filename: char[20], filesize: int)
    - S2(<u>filename</u>: char[20], filetype: char[5])

  Since S1 contains the candidate key of the original relation, no need to add an extra relation. No relation is a subset of another, so there are no redundant relations.

f) Contributor:
    - sin -> agency, cname, email, address, birthday, age
    - email -> agency, cname, address, birthday, age
    - birthday -> age

  The closures are:
  {sin}+ = sin, agency, cname, email, address, birthday, age
  {email}+ = agency, cname, email, address, birthday, age
  {birthday}+ = birthday, age

Aside from FD(sin -> agency, cname, email, address, birthday, age), the remaining FD violates 3NF. This is because their left-hand side is not a superkey, and their right-hand side contains a non-prime attribute.

- Put FDs in Standard Form:
    - sin -> agency
    - sin -> cname
    - sin -> email
    - sin -> address
    - sin -> birthday
    - sin -> age
    - email -> agency
    - email -> cname
    - email -> address
    - email -> birthday
    - email -> age
    - birthday -> age

- Minimize LHS of each FD (No Change):
    - sin -> agency
    - sin -> cname
    - sin -> email
    - sin -> address
    - sin -> birthday
    - sin -> age
    - email -> agency
    - email -> cname
    - email -> address
    - email -> birthday
    - email -> age
    - birthday -> age

- Delete Redundant FDs:
    - sin -> email
    - email -> agency
    - email -> cname
    - email -> address
    - email -> birthday
    - birthday -> age

FD(sin -> agency), FD(sin -> cname), FD(sin -> address), FD(sin -> birthday), FD(sin -> age) are dropped because you can derive the right-hand side attributes transitively through FD(email -> agency), FD(email -> cname), FD(email -> address), FD(email -> birthday), and FD(email -> age).

- Synthesis Method: Creating relations for every FD in the minimal cover
    - S1(sin: char[9], email: char[20])
    - S2(email: char[20], agency: char[20], cname: char[20], address: char[30], birthday: date)
    - S3(birthday: date, age: int)

Since S1 contains the candidate key of the original relation, no need to add an extra relation. No relation is a subset of another, so there are no redundant relations.

g) Model:
    - sin -> height

The closures are:
{sin}+ = sin, height

FD(sin -> height) is dependent on a superkey, so it is in 3NF.

- Synthesis Method: Creating relations for every FD in the minimal cover
    - S1(sin: char[9], height: int)

h) Clothes:
    - cid -> ctype, cfit, designer, price
    - designer, ctype -> price

The closures are:
{cid}+ = cid, ctype, cfit, designer, price
{designer, ctype}+ = designer, ctype, price

Aside from FD(cid -> ctype, cfit, designer, price), the remaining FD violates 3NF. This is because their left-hand side is not a superkey, and their right-hand side (price) contains a non-prime attribute.

- Put FDs in Standard Form:
    - cid -> ctype
    - cid -> cfit
    - cid -> designer
    - cid -> price
    - designer, ctype -> price

- Minimize LHS of each FD (No Change):
    - cid -> ctype

- cid -> cfit
- cid -> designer
- cid -> price
- designer, ctype -> price

- Delete Redundant FDs:
  - cid -> ctype
  - cid -> cfit
  - cid -> designer
  - designer, ctype -> price

  FD(cid -> price) is dropped because you can derive price transitively through FD(cid -> ctype), FD(cid -> designer), and FD(designer, ctype -> price).

- Synthesis Methods: Creating relations for every FD in the minimal cover
  - S1(<u>cid</u>: int, ctype: char[20], cfit: char[20], designer: char[20])
  - S2(<u>designer</u>: char[20], <u>ctype</u>: char[20], price: int)

Since S1 contains the candidate key of the original relation, no need to add an extra relation. No relation is a subset of another, so there are no redundant relations.

i) Brand:
   - bname -> established, country, owner

   The closures are:
   {bname}+ = bname, established, country, owner

   FD(bname -> established, country, owner) is dependent on a superkey, so it is in 3NF.

- Synthesis Method: Creating relations for every FD in the minimal cover
  - S1(<u>bname</u>: char[20], established: date, country: char[2], owner: char[20])

j) Wears:
   - sin, cid-> size, fit

   The closures are:
   {sin, cid}+ = sin, cid, size, fit

   FD(sin, cid-> size, fit) is dependent on a superkey, so it is in 3NF.

- Synthesis Method: Creating relations for every FD in the minimal cover
  - S1(<u>sin</u>: char[9], <u>cid</u>: int, size: char[3], fit: char[20])

Final Normalized Relations:

Publisher_Location(address: char[30], **country**: char[2], city: char[20])
- CKs: {address}

Publisher(pname: char[20], country: char[2], address: char[30], established: date)
- CKs: {pname, country}
- FKs: address (from Publisher_Location)

Magazine(ISSN: char[8], mname: char[20], genre: char[20], language: char[20], **pname**: char[20], **country**: char[2])
- CK: {ISSN}
- FK: pname and country (from Publisher)

Issue(issuenum: int, issuetitle: char[20], length: int, pdate: date, ISSN: char[8])
- CK: {issuenum, ISSN}
- FK: ISSN (from Magazine)

Article_PageRange(pagestart: int, pageend: int, pagecount: int)
- CK: {pagestart, pageend}

Article(doi: char[20], header: char[50], pagestart: int, pageend: int, **issuenum**: int, **ISSN**: char[8])
- CK: {doi}
- FK: issuenum and ISSN (from Issue), pagestart and pageend (from Article_PageRange)

Photo_FileType(filetype: char[5], filename: char[20])
- CK: {filename}

Photo(filesize: int, pid: int, filename: char[20])
- CK: {pid}
- FK: filename (from Photo_FileType)

Contains(doi: char[20], pid: int)
-CK: {doi, pid}
-FK: doi (from Article), pid (from Photo)

Birthday_Age(birthday: date, age: int)
-CK: {birthday}

Contributor_EmailDetails(agency: char[20], cname: char[20], address: char[30], email: char[20], birthday: date)
-CK: {email}
-FK: birthday (from Birthday_Age)

Contributor(sin: char[9], **email**: char[20])
-CK: {sin}
-FK: email (from Contributor_EmailDetails)

Author(sin: char[9])
-CK: {sin}
-FK: sin (from Contributor)

Photographer(sin: char[9])
-CK: {sin}
-FK: sin (from Contributor)

Model(sin: char[9], height: int)
-CK: {sin}
-FK: sin (from Contributor)

Photographed(sin: char[9], pid: int, camtype: char[20])
-CK: {sin, pid}
-FKs: sin (from Photographer), pid (from Photo)

WorkedOn(sin: char[9], issuenum: int, ISSN: char[8])
-CK: {sin, issuenum, ISSN}
-FKs: sin (from Contributor), issuenum and ISSN, (from Issue)

Clothes_Price(price: int, designer: char[20], ctype: char[20])
-CK: {designer, ctype}

Clothes(designer: char[20], ctype: char[20], cfit: char[20], cid: int)
-CK: {cid}
-FKs: designer and ctype (from Clothes_Price)

AppearsIn(cid: int, issuenum: int, ISSN: char[8])
-CK: {cid, issuenum, ISSN}
-FKs: cid (from Clothes), issuenum and ISSN (from Issue)

Wears(sin: char[9], cid: int, size: char[3], fit: char[20])
-CK: {sin, cid}
-FKs: sin (from Model), cid (from Clothes)

Brand(bname: char[20], established: date, country: char[2], owner: char[20])
-CK: {bname}

Branded(bname: char[20], cid: int)
-CK: {bname, cid}
-FKs: bname (from Brand), cid (from Clothes)

8. SQL DDL Statements:

```
CREATE TABLE Publisher_Location (
  address  VARCHAR2(30) PRIMARY KEY,
  city     VARCHAR2(20),
  country  CHAR(2) NOT NULL
);

CREATE TABLE Publisher (
  pname       VARCHAR2(20),
  country     CHAR(2),
  address     VARCHAR2(30),
  established DATE,
  PRIMARY KEY (pname, country),
  FOREIGN KEY (address) REFERENCES Publisher_Location(address)
  ON DELETE SET NULL
);

CREATE TABLE Magazine (
  ISSN     CHAR(8) PRIMARY KEY,
  mname    VARCHAR2(20),
  genre    VARCHAR2(20),
  language VARCHAR2(20),
  pname    VARCHAR2(20) NOT NULL,
  country  CHAR(2) NOT NULL,
  UNIQUE (mname, pname, country),
  FOREIGN KEY (pname, country) REFERENCES Publisher(pname, country)
  ON DELETE CASCADE
);

CREATE TABLE Issue (
  issuenum   INTEGER,
  ISSN       CHAR(8),
  issuetitle VARCHAR2(20),
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  length     INTEGER,
  pdate      DATE,
  PRIMARY KEY (issuenum, ISSN),
  UNIQUE (issuetitle, pdate, ISSN),
  FOREIGN KEY (ISSN) REFERENCES Magazine(ISSN)
  ON DELETE CASCADE
);

CREATE TABLE Article_PageRange (
  pagestart INTEGER,
  pageend   INTEGER,
  pagecount INTEGER,
  PRIMARY KEY (pagestart, pageend)
);

CREATE TABLE Article (
  doi       VARCHAR2 (20) PRIMARY KEY,
  header    VARCHAR2(50),
  pagestart INTEGER,
  pageend   INTEGER,
  issuenum  INTEGER NOT NULL,
  ISSN      CHAR(8) NOT NULL,
  FOREIGN KEY (issuenum, ISSN) REFERENCES Issue(issuenum, ISSN)
  ON DELETE CASCADE,
  FOREIGN KEY (pagestart, pageend) REFERENCES
Article_PageRange(pagestart, pageend)
  ON DELETE SET NULL
);

CREATE TABLE Photo_FileType (
  filename VARCHAR2(20) PRIMARY KEY,
  filetype CHAR(5)
);

CREATE TABLE Photo (
  pid      INTEGER PRIMARY KEY,
  filename VARCHAR2(20),
  filesize INTEGER,
  FOREIGN KEY (filename) REFERENCES Photo_FileType(filename)
  ON DELETE SET NULL
```

```
);

CREATE TABLE Contains (
  doi    VARCHAR2(20),
  pid    INTEGER,
  PRIMARY KEY (doi, pid),
  FOREIGN KEY (doi) REFERENCES Article(doi)
  ON DELETE CASCADE,
  FOREIGN KEY (pid) REFERENCES Photo(pid)
  ON DELETE CASCADE
);

CREATE TABLE Birthday_Age (
  birthday DATE PRIMARY KEY,
  age      INTEGER
);

CREATE TABLE Contributor_EmailDetails (
  email    VARCHAR2(20) PRIMARY KEY,
  agency   VARCHAR2(20),
  cname    VARCHAR2(20),
  address  VARCHAR2(30),
  birthday DATE,
  FOREIGN KEY (birthday) REFERENCES Birthday_Age(birthday)
  ON DELETE SET NULL
);

CREATE TABLE Contributor (
  sin   CHAR(9) PRIMARY KEY,
  email VARCHAR2(20) NOT NULL,
  UNIQUE (email),
  FOREIGN KEY (email) REFERENCES Contributor_EmailDetails(email)
  ON DELETE CASCADE
);

CREATE TABLE Author (
  sin CHAR(9) PRIMARY KEY,
  FOREIGN KEY (sin) REFERENCES Contributor(sin)
  ON DELETE CASCADE
);
```

```
CREATE TABLE Photographer (
  sin CHAR(9) PRIMARY KEY,
  FOREIGN KEY (sin) REFERENCES Contributor(sin)
  ON DELETE CASCADE
);

CREATE TABLE Model (
  sin    CHAR(9) PRIMARY KEY,
  height INTEGER,
  FOREIGN KEY (sin) REFERENCES Contributor(sin)
  ON DELETE CASCADE
);

CREATE TABLE Photographed (
  sin    CHAR(9),
  pid    INTEGER,
  camtype VARCHAR2(20),
  PRIMARY KEY (sin, pid),
  FOREIGN KEY (sin) REFERENCES Photographer(sin)
  ON DELETE CASCADE,
  FOREIGN KEY (pid) REFERENCES Photo(pid)
  ON DELETE CASCADE
);

CREATE TABLE WorkedOn (
  sin      CHAR(9),
  issuenum INTEGER,
  ISSN     CHAR(8),
  PRIMARY KEY (sin, issuenum, ISSN),
  FOREIGN KEY (sin) REFERENCES Contributor(sin)
  ON DELETE CASCADE,
  FOREIGN KEY (issuenum, ISSN) REFERENCES Issue(issuenum, ISSN)
  ON DELETE CASCADE
);

CREATE TABLE Clothes_Price (
  designer VARCHAR2(20),
  ctype    VARCHAR2(20),
  price    INTEGER,
```

```
  PRIMARY KEY (designer, ctype)
);

CREATE TABLE Clothes (
  cid      INTEGER PRIMARY KEY,
  ctype    VARCHAR2(20),
  cfit     VARCHAR2(20),
  designer VARCHAR2(20),
  FOREIGN KEY (designer, ctype) REFERENCES Clothes_Price(designer,
ctype)
  ON DELETE SET NULL
);

CREATE TABLE AppearsIn (
  cid      INTEGER,
  issuenum INTEGER,
  ISSN     CHAR(8),
  PRIMARY KEY (cid, issuenum, ISSN),
  FOREIGN KEY (cid) REFERENCES Clothes(cid)
  ON DELETE CASCADE,
  FOREIGN KEY (issuenum, ISSN) REFERENCES Issue(issuenum, ISSN)
  ON DELETE CASCADE
);

CREATE TABLE Wears (
  sin  CHAR(9),
  cid  INTEGER,
  size CHAR(3),
  fit  VARCHAR2(20),
  PRIMARY KEY (sin, cid),
  FOREIGN KEY (sin) REFERENCES Model(sin)
  ON DELETE CASCADE,
  FOREIGN KEY (cid) REFERENCES Clothes(cid)
  ON DELETE CASCADE
);

CREATE TABLE Brand (
  bname       VARCHAR2(20) PRIMARY KEY,
  established DATE,
  country     CHAR(2),
```

```
  owner       VARCHAR2(20)
);

CREATE TABLE Branded (
  bname VARCHAR2(20),
  cid   INTEGER,
  PRIMARY KEY (bname, cid),
  FOREIGN KEY (bname) REFERENCES Brand(bname)
  ON DELETE CASCADE,
  FOREIGN KEY (cid) REFERENCES Clothes(cid)
  ON DELETE CASCADE
);
```

- Publisher Foreign Key Integrity: We chose set null to handle address because the address is not crucial for identifying a Publisher.

- Magazine Foreign Key Integrity: We chose cascade to handle pname and country because deleting a publisher should remove all related magazines.

- Issue Foreign Key Integrity: We chose cascade to handle ISSN because deleting a magazine should remove all its issues (it is a weak entity).

- Article Foreign Key Integrity: We chose set cascade to handle issuenum and ISSN because removing an issue should remove its articles, but set null to handle pagestart, and pageend to retain article information without specific page ranges.

- Photo Foreign Key Integrity: We chose set null to handle filename because a photo's file type may exist even without the file name.

- Contains Foreign Key Integrity: We chose to cascade to handle doi and pid because removing an article or photo should remove the association.

- Contributor Email Details Foreign Key Integrity: We chose to set null to handle birthday because contributor email info can exist without a birthday.

- Contributor Foreign Key Integrity: We chose to cascade to handle email because deleting email details should remove the contributor record.

- Author, Photographer, Model Foreign Key Integrity: We chose to cascade to handle sin because deleting a contributor should remove all subtype records.

- Photographed Foreign Key Integrity: We chose to cascade to handle sin and pid because removing a photographer or photo should remove the association.

- WorkedOn Foreign Key Integrity: We chose cascade issuenum, issn, and sin because removing contributors or issues worked on should remove the association.

- Clothes Foreign Key Integrity: We chose to set null to handle designer and ctype because clothing worn or shown in a magazine may be generic without a designer, with no intentions of promoting or marketing it.

- AppearsIn Foreign Key Integrity: We chose to cascade to handle cid, issuenum, and issn because removing clothing or issues should remove their appearance records.

- Wears Foreign Key Integrity: We chose to cascade to handle sin and cid because removing a model or clothing item should remove its association.

- Branded Foreign Key Integrity: We chose to cascade to handle name and cid because deleting a Brand or Clothes should remove its association.

9. No AI tools were used.